

Introduction to Professional Practices in IT

CS4001 – Professional Practices in IT

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Internet Issues and Computer Misuse

Internet Issues

Benefits of internet:

- The benefits that the internet has brought are almost universally recognized. It has made access to all sorts of information much easier.
- It has made it much easier for people to communicate with each other, on both an individual and a group basis.
- It has simplified and speeded up many types of commercial transaction and, most importantly, these benefits have been made available to very many people, not just to a small and privileged group – although, of course, the internet is still far from being universally available, even in developed countries.
- Inevitably, a development on this scale creates its own problems.

Problems of internet availability

Every country has laws governing what can be published or publicly displayed. The following three areas are mainly covered as major problems arising due to the availability of internet:

- Defamation
- Sexual Content
- Spam
- Political and religious comment
- Depiction of violence

That are a matter of concern to everyone professionally involved in the internet, as well as to many other people. These are topics that cannot sensibly be discussed in technical terms alone. There are social, cultural and legal issues that must all be considered.

Different countries approach these issues in very different ways but the internet itself knows no boundaries.

Problems of internet availability

- Some countries, for example, consider that pictures of scantily clad women are indecent and have laws that prevent them from appearing in publications and advertisements. In other countries, such pictures are perfectly acceptable.
- In some countries, publication of material criticizing the government or the established religion is effectively forbidden, while in others it is a right guaranteed by the constitution and vigorously defended by the courts.

Availability of internet playing the role

- The coming of the internet (and satellite television) has made these differences much more apparent and much more important than they used to be.
- Since material flows across borders so easily, it is both much likelier that material that violates publication laws will come into a country and more difficult for the country to enforce its own laws.
- The roles and responsibilities of ISPs are a central element in the way these issues are addressed and we therefore start by discussing the legal framework under which ISPs operate. Then we shall look at the problems of different legal systems.

Law across National Boundaries

- Criminal law
(extradition treaty)
(extraterritorial jurisdiction)
- The international convention on cybercrime
- Civil law

Criminal Law

- Suppose a person, X, commits a criminal offence in country A and then moves to country B.
- Can country A ask that X be arrested in country B and sent back to A so that he can be put on trial?
- Or can X be prosecuted in country B for the offence committed in country A?

Criminal Law Cont.

- The answer to the first of these questions is that, provided there exists an agreement (usually called an extradition treaty) between the two countries, then in principle X can be extradited, that is, arrested and sent back to face trial in A.
- However, this can only be done under the very important provision that the offence that X is alleged to have committed in A would also be an offence in B. What is more, extradition procedures are usually extremely complex, so that attempts at extradition often fail because of procedural weaknesses.

Criminal Law Cont.

- In general, the answer to the second question is that X cannot be prosecuted in B for an offence committed in A. However, in certain cases some countries, including the UK and the USA, claim extraterritorial jurisdiction, that is the right to try citizens and other residents for crimes committed in other countries;
- In particular, this right is used to allow the prosecution of people who commit sexual offences involving children while they are abroad.
- However, the issue of extraterritoriality is much wider than this and attempts to claim extraterritorial jurisdiction make countries very unpopular.

Internet Crime

Suppose that you live in country A and on your website there you publish material that is perfectly legal and acceptable in country A, but which it is a criminal offence to publish in country B. Then you can't be prosecuted in country A and it is very unlikely that you would be extradited to country B. You might, however, be unwise to visit country B voluntarily.

Convention on Cybercrime

- In 2001, the Council of Europe approved the Convention on Cybercrime. It deals with crimes committed on the internet and other networks, including criminal copyright infringement, computer-related fraud and hacking, and child sexual abuse imagery on the internet.
- International conventions inevitably are slow to take effect. Governments sign the treaty showing that they approve of it. However, in many cases they will have to persuade their legislature to approve it and the laws necessary to implement it. This process, known as ratification.

Civil Law

- Consider the case of an ISP based in the USA with a European office in London. One of its customers is an Italian, resident in Italy, who posts an accusation about a French politician on his website (which is hosted by the ISP). The French politician complains but the ISP does nothing to remove the allegation.
- If the French politician wishes to take action, he can, in theory, do so in any of the four countries involved – England, France, Italy or the USA. His best hope of winning a court action may well be in France but there is little point in bringing an action in France unless the ISP has some sort of legal presence there. The same applies to Italy.

DEFAMATION

Defamation means

- “making statements that will damage someone’s reputation, bring them into contempt, make them disliked, and so on.”
- In England and Wales, a distinction is made between slander, which is spoken, and libel, which is written or recorded in some other way.

DEFAMATION

Possible defenses:

- a) the defendant is not the author, editor or publisher of the statement complained of;
- b) the statement complained of is substantially true;
- c) the statement complained of was a statement of opinion and it was made clear in the statement what the basis was for the opinion;
- d) the statement was published as a matter of public interest;
- e) the statement was posted on a website and the operator of the website was not responsible for posting the statement;
- f) the statement complained of was published in a peer-reviewed scientific or academic journal;
- g) the statement was a report that is protected by privilege, such as a report of proceedings at court.

SPAM

Spam is best defined as 'unsolicited email sent without the consent of the addressee and without any attempt at targeting recipients who are likely to be interested in its contents'.

Stopping Spams

There are some technical means of fighting spam, for example:

- Closing loopholes that enable spammers to use other people's computers to relay bulk messages;
- The use of machine learning and other techniques to identify suspicious features of message headers;
- The use of virus detection software to reject emails carrying viruses;
- Keeping 'stop lists' of sites that are known to send spam.

Registration

Both the USA and the UK operate successful schemes that allow individuals to register their telephone numbers as ones to which unsolicited direct marketing calls must not be made.

- In order to enforce the law, it is necessary to be able to identify reliably the source of the communication.
- Telephone operators keep records of calls showing the originator and the destination of the call; such records are needed for billing purposes.
- It is therefore easy, in most cases, to identify the source of any direct marketing call about which a consumer complains and then take the action necessary to enforce the law.

Registration

- In most cases, use of the internet is not charged on the basis of individual communications but on the basis of data transfer limits, so there is no recording of individual emails and it costs no more to send an email from Australia to the UK than it does to send an email to one's colleague in the next office.
- Furthermore, spoofing (forging the sender's address on an email) and relaying (using other people's mail servers to send your spam) are easily achieved.

COOKIES AND USER TRACKING

- Cookies are items of data that can be stored in a browser when a user accesses a site. A cookie might be used by the site that the user is visiting (e.g. to handle login information or analytics information to monitor site usage). Some sites use third-party cookies, which can be used to track users across different sites.
- The PECR (**Privacy and Electronic Communications Regulations**) also cover the use of cookies. There are requirements to inform users that cookies are being used and to provide an option to decline cookies being stored in the browser. If a user declines the use of cookies, this may have an impact on what functionality is available
- As a result of the PECR, websites that use cookies started to add statements to inform users of this fact, with an option to accept the use of cookies or decline the use of cookies.

THE COMPUTER MISUSE ACT 1990

- The Computer Misuse Act creates three new offences that can briefly be described as:
- Unauthorized access to a computer;
- Unauthorized access to a computer with intention to commit a serious crime;
- Unauthorized modification of the contents of a computer.

Section 1 of the Computer Misuse Act 1990

A person is guilty of an offence if

- he causes a computer to perform any function with intent to secure access to any program or data held in any computer;
- the access he intends to secure is unauthorized;
- he knows at the time when he causes the computer to perform the function that that is the case.

Section 2 of the Computer Misuse Act 1990

- Section 2 of the Act is concerned with gaining unauthorized access to a computer with the intention of committing a more serious offence. A blackmailer might attempt to gain unauthorized access to medical records.
- Example, in order to identify people in prominent positions who had been treated for transmitted diseases, with a view to blackmailing them. A terrorist might try to get access to a computer system for air traffic control with a view to issuing false instructions to pilots in order to cause accidents to happen.

Section 3 of the Computer Misuse Act 1990

A person is guilty of an offence if

- He does any act which causes an unauthorized modification of the contents of any computer;
- At the time when he does the act he has the requisite intent and the requisite knowledge.

Section 3 of the Computer Misuse Act 1990 Cont.

The requisite intent is an intent to cause a modification of the contents of any computer and by so doing

- to impair the operation of any computer;
- to prevent or hinder access to any program or data held in any computer;
- to impair the operation of any such program or the reliability of any such data.

Section 3 of the Computer Misuse Act 1990 Cont.

It is the offence created by Section 3 that gives the Act its power. For example, it makes each of the following a criminal offence:

- Intentionally spreading a virus, worm, or other pest;
- Encrypting a company's data files and demanding a ransom for revealing the key required to decrypt it;
- Concealed redirection of browser home pages;
- Implanting premium rate dialers (that is, programs that replace the normal dial-up code for the computer with the code for a premium rate service).

Computer Fraud

The Law Commission defined computer fraud as: conduct that involves the manipulation of a computer, by whatever method, dishonestly obtain money, property, or some other advantage of value, or to cause loss.

Computer fraud involves manipulating a computer dishonestly in order to obtain

- money,
- property,
- services,
- to cause loss.

Fraud Techniques

- Most of the techniques that are used are much older than computers. Such tricks as
- placing fictitious employees on the payroll or
- setting up false supplier accounts and creating spurious invoices

are still the commonest type of fraud as they were before computers appeared.

Computer Crime

Alternatively referred to as cyber crime, crime, electronic crime, or hi-tech crime. Computer crimes an act performed by a knowledgeable computer user, sometimes referred to as a hacker that illegally browses or steals a company's or individual's private information. In some cases, this person or group of individuals may be malicious and destroy or otherwise corrupt the computer or data files.